

## TECHNIQUE MASTERING EXERCISES

### N INTEGRATION BY INSPECTION

ii HAVE SIMPLIFY THE ANSWERS, BUT YOU DO NOT HAVE TO SIMPLYFY IT.

$$1. \int e^{5x} dx = \frac{1}{5} \int 5e^{5x} dx = \frac{1}{5} e^{5x} + c$$

$$2. \int x^8 dx = \frac{1}{9} x^9 + c$$

$$3. \int \frac{1}{x^6} dx = \int x^{-6} dx = \frac{x^{-5}}{-5} + c = \frac{1}{5x^5} + c$$

$$4. \int t^{\frac{5}{7}} dt = \frac{t^{\frac{12}{7}}}{\frac{12}{7}} + c = \frac{7}{12} t^{\frac{12}{7}} + c$$

$$5. \int \sqrt[3]{x^2} dx = \frac{x^{\frac{5}{3}}}{\frac{5}{3}} + c = \frac{3}{5} x^{\frac{5}{3}} + c$$

$$6. \int \frac{4}{\sqrt{x}} dx = \int 4x^{-\frac{1}{2}} dx = 4 \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c = 8\sqrt{x} + c$$

$$7. \int \frac{1}{2\theta^3} d\theta = \frac{1}{2} \int \theta^{-3} d\theta = \frac{1}{2} \frac{\theta^{-2}}{-2} + c = \frac{-1}{4\theta^2} + c$$

$$8. \int x^3 \sqrt{x} dx = \int x^{\frac{7}{2}} dx = \frac{x^{\frac{9}{2}}}{\frac{9}{2}} + c = \frac{2}{9} x^{\frac{9}{2}} + c$$

$$9. \int (x^3 - 2x + 7) dx = \frac{1}{4} x^4 - x^2 + 7x + c$$

$$10. \int (x^{-3} + \sqrt{x} - 3x^{1/4} + x^2) dx = \frac{x^{-2}}{-2} + \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - 3 \frac{x^{\frac{5}{4}}}{\frac{5}{4}} + \frac{x^3}{3} + c = -\frac{1}{2x^2} + \frac{2}{3} \sqrt{x^3} - \frac{12}{5} \sqrt[4]{x^5} +$$

$$11. \int (y^{2/3} - 4y^{1/5} + 4) dy = \frac{y^{\frac{5}{3}}}{\frac{5}{3}} - 4 \frac{y^{\frac{6}{5}}}{\frac{6}{5}} + 4y + c = \frac{3}{5} \sqrt[5]{y^5} - \frac{10}{3} \sqrt[6]{y^6} + 4y + c$$

$$12. \int \left( \frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x} \right) dx = 7 \frac{x^{\frac{1}{4}}}{\frac{1}{4}} - \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + 4 \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + c = 28\sqrt[4]{x} - \frac{3}{4} \sqrt[3]{x^4} + \frac{8}{3} \sqrt{x^3} + c$$

$$13. \int x(1+x^3) dx = \int (x+x^4) dx = \frac{x^2}{2} + \frac{x^5}{5} + c = \frac{1}{2} x^2 + \frac{1}{5} x^5 + c$$

$$14. \int (1+u^2)(2-u) du = \int (2-u+2u^2-u^3) du = 2u - \frac{u^2}{2} + 2 \frac{u^3}{3} - \frac{u^4}{4} + c = -\frac{1}{4} u^4 + \frac{2}{3} u^3 - \frac{1}{2}$$

$$15. \int x^{1/3} (2-x)^2 dx = \int \left( 4x^{\frac{1}{3}} - 4x^{\frac{4}{3}} + x^{\frac{7}{3}} \right) dx = 4 \frac{x^{\frac{4}{3}}}{\frac{4}{3}} - 4 \frac{x^{\frac{7}{3}}}{\frac{7}{3}} + \frac{x^{\frac{10}{3}}}{\frac{10}{3}} + c = 3\sqrt[3]{x^4} - \frac{12}{7} \sqrt[3]{x^7} + \frac{3}{10}$$

$$16. \int \left( \frac{1}{t^2} - \cos t \right) dt = \frac{t^{-1}}{-1} - \sin t + c = -\frac{1}{t} - \sin t + c$$

$$17. \int (3 \sin x - 4 \cos x) dx = \int (-3(-\sin x) - 4 \cos x) dx = -3 \cos x - 4 \sin x + c$$

18.  $\int e^{-x} dx = -\int -e^{-x} dx = -e^{-x} + c = \frac{-1}{e^x} + c$
19.  $\int 2xe^{x^2} dx = e^{x^2} + c$
20.  $\int \frac{1}{2x} dx = \frac{1}{2} \int \frac{1}{x} dx = \frac{1}{2} \ln|x| + c$
21.  $\int e^{3t} dt = \frac{1}{3} \int 3e^{3t} dt = \frac{1}{3} e^{3t} + c$
22.  $\int \frac{3}{x} dx = 3 \int \frac{1}{x} dx = 3 \ln|x| + c$
23.  $\int \frac{1}{2x+1} dx = \frac{1}{2} \int \frac{2}{2x+1} dx = \frac{1}{2} \ln|2x+1| + c$
24.  $\int \sinh 4x dx = \frac{1}{4} \cosh 4x + c$
25.  $\int \sinh (4x+6) dx = \frac{1}{4} \cosh(4x+6) + c$
26.  $\int \cosh (2x+3) dx = \frac{1}{2} \sinh(2x+3) + c$
27.  $\int \tanh 3x dx = \frac{1}{3} \int \frac{3 \sinh 3x}{\cosh 3x} dx = \frac{1}{3} \ln|\cosh 3x| + c$
28.  $\int \frac{x^2}{x^3-4} dx = \frac{1}{3} \int \frac{3x^2}{x^3-4} dx = \frac{1}{3} \ln|x^3-4| + c$
29.  $\int \frac{5u^4}{u^5+1} du = \ln|u^5+1| + c$
30.  $\int \frac{t+1}{t} dt = \int \left( \frac{t}{t} + \frac{1}{t} \right) dt = \int \left( 1 + \frac{1}{t} \right) dt = t + \ln|t| + c$
31.  $\int \frac{x^3}{x^2+1} dx = \int \frac{x(x^2+1)-x}{x^2+1} dx = \int \frac{x(x^2+1)}{x^2+1} + \frac{-x}{x^2+1} dx = \int x + \frac{-x}{x^2+1} dx = \int x dx - \frac{1}{2} \int$
32.  $\int x^2(2-x^3)^4 dx = \frac{1}{-3} \int (2-x^3)^4(-3x^2) dx = \frac{(2-x^3)^5}{-15} + c$
33.  $\int xe^{2x^2} dx = \frac{1}{4} \int 4xe^{2x^2} dx = \frac{1}{4} e^{2x^2} + c$
34.  $\int \sin \theta (\cos \theta - 3)^3 d\theta = \int (\cos \theta - 3)^3 (\sin \theta) d\theta = \frac{(\cos \theta - 3)^4}{4} + c$
35.  $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = -2 \int (-\sin \sqrt{x}) \left( \frac{1}{2\sqrt{x}} \right) dx = -2 \cos \sqrt{x} + c$
36.  $\int \frac{\sin \theta}{1+\cos \theta} d\theta = -\int \frac{-\sin \theta}{1+\cos \theta} d\theta = -\ln|1+\cos \theta| + c$
37.  $\int \tan \theta d\theta = -\int \frac{-\sin \theta}{\cos \theta} d\theta = -\ln|\cos \theta| + c$

$$38. \int \frac{1}{x \ln x} dx = \int (\ln x)^{-1} \left( \frac{1}{x} \right) dx = \ln|\ln x| + c$$

$$39. \int \frac{\ln y}{y} dy = \int (\ln y)^1 \left( \frac{1}{y} \right) dy = \frac{1}{2} \ln^2 y + c$$

$$40. \int (x+1) \sqrt{1-2x} dx$$

$$\text{Let } \sqrt{1-2x} = u \text{ then } (\sqrt{1-2x})^2 = (u)^2 \Rightarrow 1-2x = u^2 \Rightarrow x = \frac{1-u^2}{2} \Rightarrow \frac{dx}{du} = -u$$

$$\int (x+1) \sqrt{1-2x} dx = \int \left( \frac{1-u^2}{2} + 1 \right) u (-u du) = \frac{-1}{2} \int (3-u^2) u^2 du = \frac{-1}{2} \int (3u^2 - u^4) du$$

$$= \frac{-1}{2} \left( 3 \frac{u^3}{3} - \frac{u^5}{5} \right) + c = -\frac{1}{2} u^3 + \frac{1}{10} u^5 + c = -\frac{1}{2} (\sqrt{1-2x})^3 + \frac{1}{10} (\sqrt{1-2x})^5 + c$$